

Declaration of Generative AI Use

Unit: FIT3143 Parallel Computing, Semester 2 2026

Assessment: Lab 2, Message Passing Interface (Week 8)

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Date: 8 September 2026 [last touched]

Declaration

We used generative AI during the preparation period for this lab, as allowed under item 9 of the assessment specification. We did not use any AI tool during the presentation or the Q&A, as required by item 10.

Tool used: Claude (Anthropic), during the week of 1 to 8 September 2026.

What it was used for, by task:

- Task 1 and Task 2 (Open MPI and hybrid code). Taabish drafted both programs.

AI was used to review the drafts, and it identified one real defect: task2.c called memcopy without including <string.h>, which made the file fail to compile under Apple clang. The one line include was added; nothing else in the search, the partitioning or the collectives was changed by AI.

- Task 3 (performance evaluation). AI was used to help design the measurement experiment, in particular the decision to instrument separate copies of the two programs (task1_instr.c, task2_instr.c, serial_instr.c) rather than time the submitted code, and the decision to take the maximum phase time across ranks rather than rank 0's own. It was also used to draft run_benchmarks.sh and make_graphs.py, and to check the Amdahl algebra. The measurement design, the choice of baseline, and every number in the results are our own; the numbers were produced by running the code on our own machine and can be reproduced by running ./run_benchmarks.sh.

- Task 4 (slides). AI was used to draft the slide text and to check the deck against the Lab 2 marking rubric for coverage of all seven required graphs, and to write make_diagrams.py, which draws the three explanatory diagrams on slides 3, 4 and 8. All graphs were generated from our own measured CSV data, and the diagrams are drawings of our own code and our own measured numbers.

Content in the slides and in Task3_Performance_Evaluation.md that is AI drafted synthesis rather than a direct report of our own measurements is written as such where it appears.

Prompt records

The full prompt and response record is attached separately as AI_Prompt_Records.pdf, as required by item 9.

Signatures

Name	Student ID
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We declare that the above is a complete and accurate account of generative AI use in this assessment, and that we understand and can explain all of the submitted work.